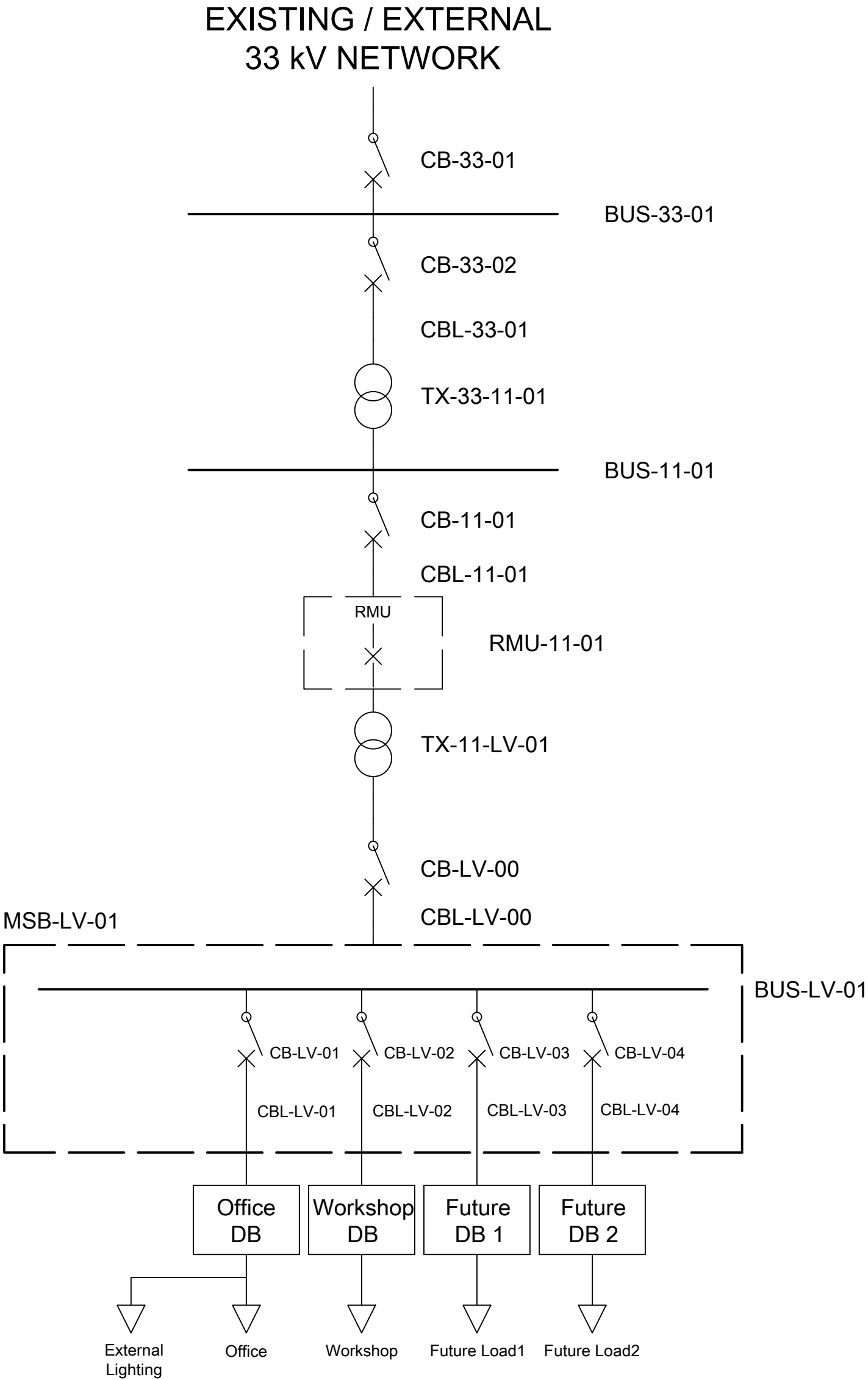


Document Control	
Field	Value
Project	Green Valley Conceptual Mining Site Office & Workshop Electrical Distribution Design
Document	Electrical Single Line Diagram
Status	Preliminary
Revision	Rev 01
Prepared by	Eason Zhu
Date	17-Sep-26
Design Case	Rev 01 Preliminary Electrical Distribution Design

Version Control		
Revision	Description of Changes	Date
Rev 00	Initial preliminary issue	—
Rev 01	Updated electrical distribution topology, feeder arrangement and equipment ratings based on revised maximum demand and preliminary transformer sizing.	17-Sep-26



- NOTES:
- 1. Conceptual electrical design developed for portfolio demonstration.
 - 2. Equipment ratings marked TBD are subject to detailed design.
 - 3. Maximum demand analysis completed in load input analysis Rev 01
 - 4. Cable sizing and protection studies are in progress.
 - 5. Planned studies include power factory load flow, short-circuit analysis, motor starting assessment and preliminary protection coordination.
 - 6. Conceptual design only. Not for construction.

A. PRELIMINARY MAJOR EQUIPMENT SCHEDULE

ID	Description	Voltage	Rating / Size	Status
BUS-33-01	33 kV Main Bus	33 kV	Bus rating TBD	Preliminary
TX-33-11-01	33/11 kV Transformer	33/11 kV	Rating TBD / upstream system basis	Preliminary
BUS-11-01	11 kV Main Bus	11 kV	Bus rating TBD	Preliminary
RMU-11-01	Green Valley Ring Main Unit	11 kV	Rating TBD	Preliminary
TX-11-LV-01	Green Valley Distribution Transformer	11/0.415 kV	100 kVA, Dyn11	Preliminary
MSB-LV-01	415 V Main Switchboard	415 V	Main bus rating TBD	Preliminary
BUS-LV-01	415 V MSB Bus	415 V	Bus rating TBD	Preliminary
DB-OF-01	Office Distribution Board	415/230 V	Rating TBD	Preliminary
DB-WS-01	Workshop Distribution Board	415/230 V	Rating TBD	Preliminary
DB-FUT-01	Future Distribution Board 1	415/230 V	Future provision	Reserved
DB-FUT-02	Future Distribution Board 2	415/230 V	Future provision	Reserved

B. PRELIMINARY FEEDER & SWITCHGEAR SCHEDULE

ID	Description / Service	Voltage	Rating / Size	Status
CB-33-01	33 kV Incoming Circuit Breaker	33 kV	TBD	Preliminary
CB-33-02	33 kV Transformer Feeder Circuit Breaker	33 kV	TBD	Preliminary
CBL-33-01	33 kV Transformer Feeder	33 kV	Cable size TBD	Future/Preliminary
CB-11-01	Green Valley 11 kV Feeder Circuit Breaker	11 kV	TBD	Preliminary
CBL-11-01	Green Valley 11 kV Feeder	11 kV	Cable size TBD	Size Pending
CB-LV-00	MSB Main Incomer	415 V	Rating TBD	Preliminary
CBL-LV-00	Transformer LV-MSB Connection	415 V	Size TBD	Preliminary
CB-LV-01	Office DB Feeder Circuit Breaker	415 V	Rating TBD	Sizing in Progress
CBL-LV-01	Office DB Feeder	415 V	IB≈30.7A; cable size TBD	Sizing in Progress
CB-LV-02	Workshop DB Feeder Circuit Breaker	415 V	Rating TBD	Sizing in Progress
CBL-LV-02	Workshop DB Feeder	415 V	IB≈78.5A; cable size TBD	Sizing in Progress
CB-LV-03	Future Feeder 1 Circuit Breaker	415 V	TBD	Reserved
CBL-LV-03	Future DB 1 Feeder	415 V	Not selected	Reserved
CB-LV-04	Future Feeder 2 Circuit Breaker	415 V	TBD	Reserved
CBL-LV-04	Future DB 2 Feeder	415 V	Not selected	Reserved